# Parth Sachin Deshmukh

parthdeshmukh321@gmail.com | (979) 492-9902 | LinkedIn | GitHub | HuggingFace | Portfolio

## Education

**Texas A&M University** — M.S. Computer Science — GPA: **4.0/4.0** May 2027 (expected)
*Coursework: Distributed Systems (in progress), Analysis of Algorithms, Artificial Intelligence, Data Mining*

**Vishwakarma Institute of Information Technology (VIIT)** — B.Tech. IT — GPA: **9.12/10.00** May 2024

## Skills

**Languages:** Python, C++, TypeScript, JavaScript, SQL, Java, Go
**Backend & APIs:** FastAPI, REST APIs, Flask, Pydantic, API security (API keys, RLS), React, LangGraph, MCP, OpenAI SDK
**Distributed & Data:** Kafka, Temporal, PostgreSQL, MySQL, SQLite, Neo4j, Qdrant, Pinecone, data modeling
**Cloud & DevOps:** AWS (EC2, ECR, CloudWatch), Docker, GitHub Actions (CI/CD), Linux, Git, Kubernetes
**Testing & Observability:** unit testing (pytest, GoogleTest), ASan/UBSan, debugging, performance optimization, OpenTelemetry

## Experience

**ML/Data Research Intern** — Indian Institute of Technology (IIT) Bombay Feb 2024 – Sep 2024

- Gathered requirements from **20+** faculty and built the analytics web app they used for reporting (JavaScript, AJAX, PHP, MySQL).
- Indexed MySQL queries across **500+** report configs and added async loading, cutting report latency **~40–80%**.
- Wrote a Figma-to-HTML parser behind **100+** reusable reporting themes; trained faculty and turned their feedback into fixes.

## Projects

**RAG-Powered Code Assistant** *GitHub* Mar 2026

- Built a code assistant over **5,586** Python, pandas and NumPy doc pages, served by an API-key-secured streaming FastAPI endpoint.
- Implemented BM25 on a hand-written inverted index + rank fusion with dense search: recall@5 **0.88** vs **0.55** on **220** auto-generated questions.
- Moved the tokenizer hot path to pybind11 C++ (CMake), benchmarked **4.9x** faster than pure Python on **4.46M** tokens.
- Shipped **106** tests at **93%** coverage and a GitHub Actions workflow that deploys the Docker app to Hugging Face Spaces.

**Realtime Multi-Agent Knowledge Graph** *GitHub* Jun 2026 – Present

- Ran each Kafka speech-to-text segment as a Temporal workflow through **4** LangGraph agents, with at most **5** running at once.
- Gated Neo4j writes behind an MCP commit tool (approved entities only, MERGE upserts) with **240** s activity timeouts.
- Served Neo4j Cypher + Qdrant retrieval through FastAPI and traced every stage with OpenTelemetry + Langfuse.

**BuildFlow** — C++17 Build and Test Scheduler Oct 2026

- Wrote a C++17 DAG build scheduler (Kahn's algorithm + priority queue) on **4** std::thread workers: **3.07x** vs 1 on **32** simulated jobs.
- Modeled the best case as max(critical path, work/4) = **3.45** s; the run reached **72%** of it, with dependency stalls as the gap.
- Flagged **8/8** injected regressions with the right failure category; **16** GoogleTest + **6** pytest tests, clean under ASan/UBSan.

**Career Ops** — Job Ingest & Resume Tooling Jun 2026 – Present

- Integrated **10+** job sources with bounded concurrency and 429/5xx backoff; deduped postings with Pydantic models and SQLite.
- Built with Cursor and Claude under AGENTS.md rules; every AI-written diff must pass **800+** pytest tests and ATS parse gates.

## Publications

- **First Author**, Springer 2024: *Automated GUI Testing to Enhance UX: A Survey of the State of the Art*
- Co-Author, IEEE 2024: *Increasing Donation Transparency in Disaster Relief: A Blockchain-based Solution*